

Euclid's Elements — Book I

Proposition 9

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

Bisecting a given rectilinear angle

1 Introduction

Proposition I.9 constructs a bisector of a given nondegenerate rectilinear angle. Euclid's classical proof and the current Lean reconstruction are both synthetic, but they use different auxiliary configurations.

Euclid chooses equal segments on the two arms of the angle, constructs an equilateral triangle on the segment joining their endpoints, and invokes Proposition I.8. The current production theorem instead lays off a segment on one original ray, takes the midpoint of the segment joining that point to the other original endpoint, and applies the general Hilbert SSS theorem.

This distinction is mathematically significant. The formal proof does not call the Book I theorems I.1, I.3, or I.8. It uses the reusable Hilbert congruence and midpoint infrastructure directly.

2 The Classical Configuration

Let $\angle BAC$ be the given rectilinear angle. The hypothesis is nondegeneracy: the points A, B, C are not collinear. The objective is to construct a ray from A that divides the angle into two congruent angles.

2.1 Euclid's construction

Choose a point D on the side AB . By Proposition I.3 cut off on the other side a segment AE satisfying

$$AD \cong AE.$$

Join D to E , construct the equilateral triangle DEF on DE by Proposition I.1, and join A to F .

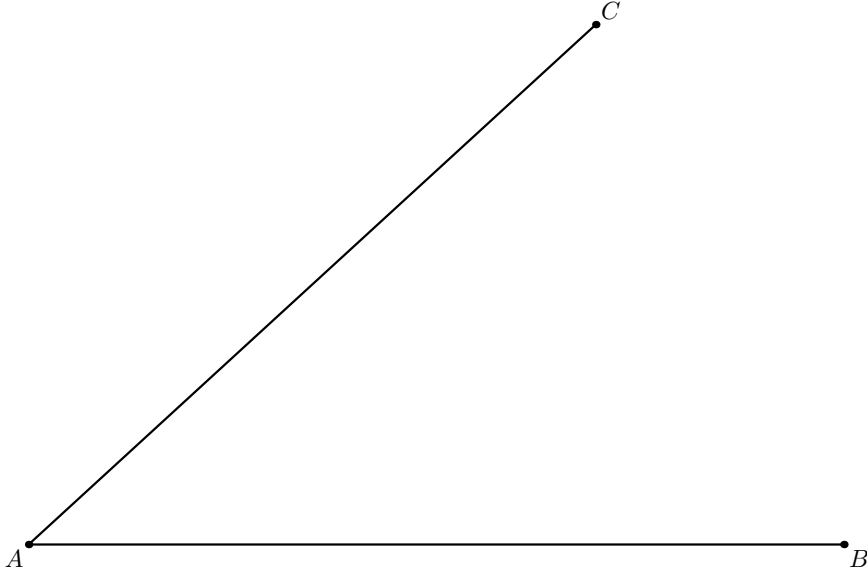


Figure 1: The given nondegenerate angle $\angle BAC$.

Since

$$AD \cong AE, \quad DF \cong EF, \quad AF \cong AF,$$

Proposition I.8 gives

$$\angle DAF \cong \angle FAE.$$

Because D and E lie on the original rays AB and AC , the ray AF bisects the given angle.

3 Statement

Theorem 1 (Euclid I.9). *Let A, B, C be noncollinear. Then there exists a point M such that*

$$\angle BAM \cong \angle CAM.$$

Thus the ray determined by A and M bisects the angle whose arms are the rays AB and AC .

The order of the last two letters in the second displayed angle follows the current Lean endpoint literally:

```
Geo.AngleCongruent B A M C A M
```

The project's angle object is symmetric in its two arms, so this is the same geometric bisection normally written $\angle BAM \cong \angle MAC$.

Diagrammatic audit. The audited chapter uses three geometric figures. Figures 1 and 2 retain the two classical states already present in Book1R. The old non-geometric flowchart for the Hilbert reconstruction is replaced by Figure 3, which records the actual stable formal configuration. No additional panel is needed: the proof has no geometric case split, and the final midpoint configuration contains the earlier segment-layoff state without ambiguity.

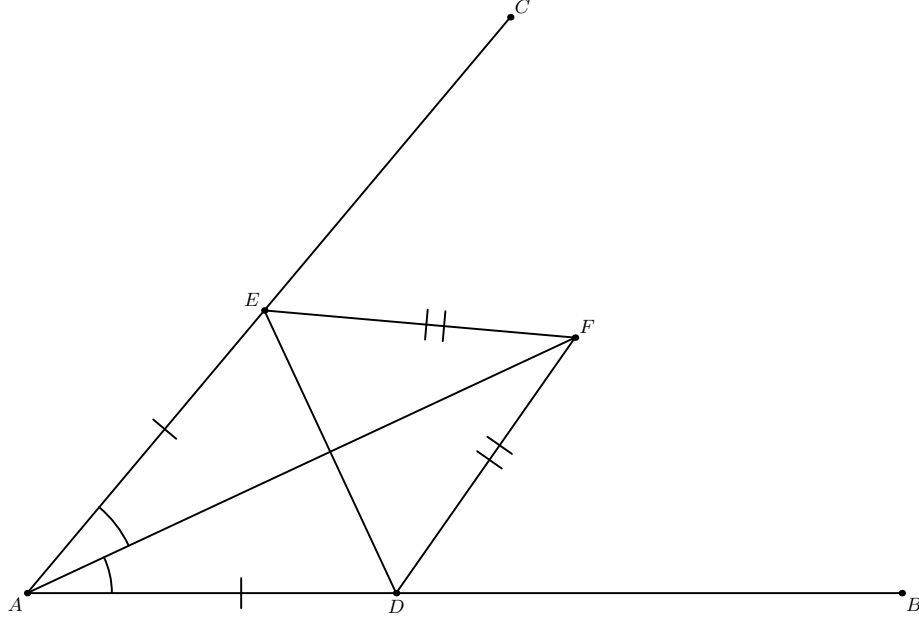


Figure 2: Euclid’s construction for Proposition I.9. The points D, E satisfy $AD \cong AE$, and DEF is equilateral. With AF common, SSS for $\triangle ADF$ and $\triangle AEF$ gives equality of the two angles at A .

4 Classical Proof

The classical dependency chain is

$$\text{I.3} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{I.9}.$$

Proposition I.3 supplies equal auxiliary segments on the two arms, Proposition I.1 supplies the equilateral triangle, and Proposition I.8 converts three side congruences into the required equality of angles.

This chain is historical, not the direct call graph of the current Lean implementation.

5 Proof Architecture of the Reconstruction

The current production theorem is not a thin wrapper. It performs the following proposition-specific construction:

$$\begin{aligned}
 &A, B, C \text{ noncollinear} \\
 &\quad \downarrow \\
 &D \text{ on ray } AB, \quad AD \cong AC \\
 &\quad \downarrow \\
 &M \text{ midpoint of } DC \\
 &\quad \downarrow \\
 &\triangle ADM \text{ and } \triangle ACM \text{ satisfy SSS} \\
 &\quad \downarrow \\
 &\angle DAM \cong \angle CAM \\
 &\downarrow \text{ same-ray transport } AD \rightsquigarrow AB \\
 &\angle BAM \cong \angle CAM.
 \end{aligned}$$

The midpoint call is deliberately treated here as a reusable theorem-level component. Unlike

Proposition I.10, I.9 already has a substantial local geometric construction. Recursively expanding the internal proof of `HilbertMidpointExists` inside this chapter would duplicate a general midpoint theorem rather than clarify the proposition-specific geometry.

6 Hilbert Reconstruction

6.1 Lay off a congruent segment on the original ray

From noncollinearity the proof derives $A \neq B$. Hilbert III.1 is then used in the form

```
HilbertCongruence.segment_construction
(Geo := Geo) A C A B hAB
```

to obtain a point D satisfying

$$\text{SameRay}(A; B, D), \quad AD \cong AC.$$

The same-ray information is essential: it is what later allows the formal angle $\angle DAM$ to be transported back to the original arm AB .

6.2 Construct the midpoint of the cross-segment

The proof establishes that A, D, C are noncollinear and therefore $D \neq C$. It then invokes

```
HilbertMidpointExists Geo D C hDC
```

and obtains M with

$$D - M - C, \quad DM \cong MC.$$

The strict betweenness is not decorative bookkeeping. It supplies both the position of M on DC and the distinctness facts needed to prove that $\triangle ADM$ is nondegenerate.

6.3 SSS and the final ray transport

The midpoint congruence is reversed on its second segment so that the SSS arguments have the exact orientation expected by the theorem. Reflexivity supplies

$$AM \cong AM.$$

The current call is

```
have hSSS :=
  HilbertSSS
    Geo
    A D M
    A C M
    hADM
    hAD_AC
    hDM_CM
    hAM
```

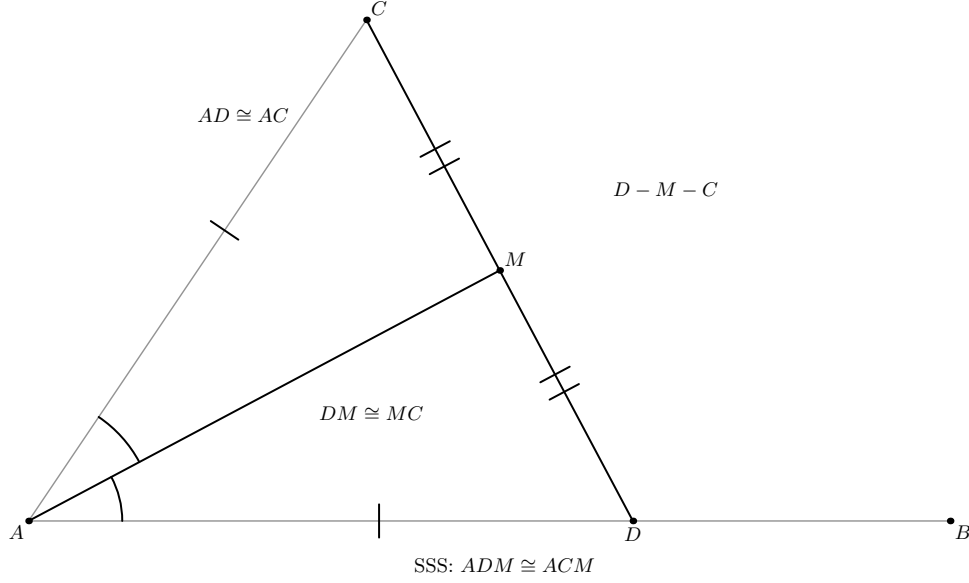


Figure 3: The formal Hilbert configuration used by the current `euclid_proposition_9`. Segment construction gives D on the original ray AB with $AD \cong AC$. The reusable midpoint theorem gives $D - M - C$ and $DM \cong MC$. Together with the common side AM , these are the three side congruences for SSS on $\triangle ADM$ and $\triangle ACM$.

The current version of `HilbertSSS` requires noncollinearity only for the first triangle; noncollinearity of the second is derived internally from the three side congruences. This is a small but concrete correction to older documentation that displayed an extra noncollinearity argument in the call.

From SSS the proof extracts

$$\angle DAM \cong \angle CAM.$$

Finally, symmetry of the same-ray relation gives D and B on the same ray from A , and `hilbert_angle_eq_of_sameRay_first` identifies $\angle DAM$ with $\angle BAM$. This yields the public conclusion.

7 Lean Formalization

The production source imports only

```
import CGJteamLab.HilbertInterface
```

and the public theorem has the form

```
theorem euclid_proposition_9
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not Collinear Geo A B C) :
  exists M : Geo.Point,
    Geo.AngleCongruent B A M C A M := by
  ...
```

There are no proposition-local helper theorems and no geometric case splits. The proof is

nevertheless substantive: it contains segment construction, noncollinearity propagation, midpoint construction, SSS, and final same-ray angle transport.

Source inspection of the current `Proposition09.lean` finds no `sorry` and no `proposition-local axiom`. This documentation audit did not run a fresh `lake build` or `#print axioms`, so it makes no stronger compilation or axiomatic-status claim.

8 Relationship with Earlier Propositions

8.1 Proposition I.3

I.3 is part of Euclid’s historical construction: it cuts off on one arm a segment equal to the chosen segment on the other arm. The current Lean proof does not call `euclid_proposition_3`; it uses Hilbert III.1 segment construction directly.

8.2 Proposition I.1

Euclid constructs the equilateral triangle DEF . No equilateral triangle is constructed in the formal proof, and `euclid_proposition_1` is not a direct dependency of `euclid_proposition_9`.

8.3 Proposition I.8

I.8 is the classical SSS step. The formal proof uses the reusable theorem `HilbertSSS` directly instead of calling `euclid_proposition_8`. This is an API distinction, not a change from synthetic to numerical geometry.

8.4 Transition to Proposition I.10

Classically, I.10 immediately reuses I.9 to bisect the apex angle of an equilateral triangle. In the current Lean development, however, I.10 is exposed through the independent midpoint theorem rather than by calling `euclid_proposition_9`. The historical dependency and the formal call graph therefore diverge at exactly this point.

9 Hilbert and Book Zero Components Used

9.1 Hilbert III.1: segment construction

The construction of D is a direct use of the congruence axiom for laying off a segment on a prescribed ray.

9.2 Strict midpoint infrastructure

`HilbertMidpointExists` returns the strict midpoint predicate, hence both

$$D - M - C \quad \text{and} \quad DM \cong MC.$$

The theorem is reused as an interface component; its internal neutral proof is not duplicated inside the proposition-specific diagram.

9.3 Hilbert SSS

The theorem `HilbertSSS` supplies triangle congruence from the three side congruences. Its current API derives noncollinearity of the second triangle internally.

9.4 Same-ray angle transport

The final result from SSS is expressed using the constructed arm AD . `hilbert_sameRay_symm` and `hilbert_angle_eq_of_sameRay_first` replace it by the original arm AB without changing the geometric angle.

No Book Zero theorem is called directly by `Proposition09.lean`. The file works through the Hilbert interface imported above.

10 Formalization Notes

The principal documentary issue in I.9 is not a missing classical step but a misleading representation of the formal proof. A flowchart saying “segment construction – midpoint – SSS” records theorem calls, not the geometry that makes those calls meaningful. The audited Figure 3 instead shows the actual symmetric configuration A, D, M, C .

A second correction concerns the endpoint. The current theorem returns only an angle-congruence witness M ; it does not package an explicit “interior-of-angle” or side-of-line predicate in its conclusion. The proof contains strict betweenness $D - M - C$ and same-ray information for B, D , which are exactly the positional facts used by the argument. Documentation should not strengthen this to a formal side-of-line conclusion that the theorem does not state.

Finally, the midpoint theorem is a reusable lower-layer result. Expanding its entire internal construction here would obscure modularity and duplicate the midpoint audit. I.9 should document how the midpoint is *used*: it creates the symmetric cross-segment that makes the SSS comparison possible.

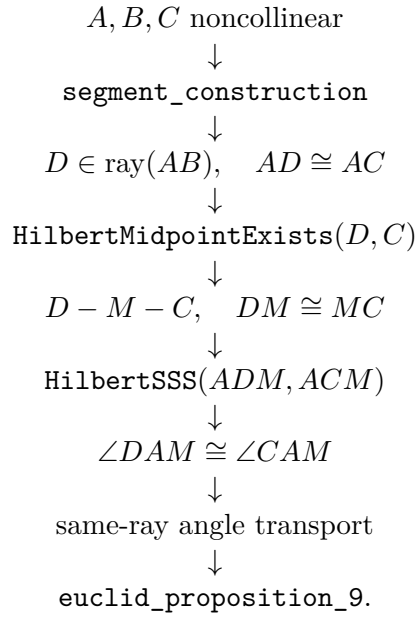
11 Dependency Summary

The classical and current formal dependency structures are:

Classical:

$$\text{I.3} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{I.9}.$$

Current formal reconstruction:



No numerical angle measure, coordinate system, or Euclidean parallel axiom is introduced by this proposition-level reconstruction.

12 Conclusion

Proposition I.9 is the first Book I construction in this sequence for which the current Hilbert proof has a visibly different auxiliary figure from Euclid’s proof. Euclid obtains symmetry from an equilateral triangle and I.8; Lean obtains it from segment layoff, a strict midpoint, and the reusable Hilbert SSS theorem.

The audited documentation now shows both pictures rather than replacing the formal geometry by a theorem-call flowchart. The formal endpoint is also recorded exactly: the constructed ray gives

$$\angle BAM \cong \angle CAM,$$

with the original arm AB recovered by same-ray transport.